**QPCODE** **Reg.No : .....**

**Name : .....**

# MAHATMA GANDHI UNIVERSITY, KOTTAYAM
## MGU-UGP (HONOURS) EXAMINATION MONTH 2025
## FIFTH SEMESTER
## DSE- MG5DSEBIF304- Genetic Engineering
(2024 ADMISSION ONWARDS)

**Duration: 2hrs** **Maximum Marks: 70**

***Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)***

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

## Part A

Short Answer Questions

Answer any **5** questions

Each question carries **2** marks

1. Define genetic engineering. [K] / [CO1]

2. What is gene knockout? [K] / [CO2]

3. Define gene therapy [K] / [CO3]

| | | |
|---|---|---|
| 4. | What is a genetically modified organism (GMO)? | [K] / [CO4] |
| 5. | What is informed consent in genetic engineering research? | [K] / [CO5] |
| 6. | What is epigenome editing? | [K] / [CO6] |
| 7. | Name one emerging application of genetic engineering.. | [K] / [CO6] |

**[2x5 = 10]**

## Part B

Short Essay Questions

Answer any **6** questions

Each question carries **5** marks

| | | |
|---|---|---|
| 8. | Explain the basic scheme of the genetic engineering process. | [U] / [CO1] |
| 9. | How can gene knockout be used to study the function of a particular gene? | [A] / [CO2] |
| 10. | Differentiate between somatic and germline gene therapy. | [U] / [CO3] |
| 11. | Mention two advantages of genetically modified crops in agriculture. | [U] / [CO4] |
| 12. | State any two ethical concerns associated with genetic engineering. | [U] / [CO5] |
| 13. | Mention two recent advances in genetic engineering that demonstrate its future potential. | [A] / [CO6] |

14. Explain the principles and applications of restriction enzymes and DNA ligase in recombinant DNA technology. [U] / [CO1]

15. A researcher wants to increase the production of a specific protein in *E. coli*. Suggest an appropriate genetic engineering strategy and explain the major steps involved. [A] / [CO2]

**[5x6 = 30]**

## Part C

Long Essay Questions

Answer any **3** questions

Each question carries **10** marks

16. Explain the major approaches used in gene therapy and discuss their applications in human disease management. [An] / [CO3]

17. Evaluate the potential benefits and risks of genetically modified crops with respect to agriculture and the environment. [E] / [CO4]

18. Discuss the ethical issues related to privacy, informed consent and germline modification in genetic engineering. [E] / [CO5]

19. Describe any two emerging advances in genetic engineering and explain how they may contribute to future research and biotechnology. [E] / [CO6]

20. **a)** Explain the major steps involved in recombinant DNA technology. [A] / [CO1]
**b)** Explain the significance of gene editing in modern genetic engineering.

**[10x3 = 30]**